

National Coverage Determination (NCD)

Home Oxygen Use to Treat Cluster Headache (CH)

240.2.2

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Tracking Information

Publication Number

100-3

Manual Section Number

240.2.2

Manual Section Title

Home Oxygen Use to Treat Cluster Headache (CH)

Version Number

1

Effective Date of this Version

01/04/2011

Ending Effective Date of this Version

09/27/2021

Implementation Date

02/15/2011

Description Information

Benefit Category

Durable Medical Equipment

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Cluster headache (CH), as described in Harrison's "Principles of Internal Medicine" 16th edition, is an episodic (most common), or chronic unilateral headache syndrome that begins with one to three short-lived headaches per day over many weeks followed by a period of remission. There may be a regular recurrence in the vast majority of attacks. When it becomes chronic, it is characterized by the absence of sustained periods of remission. Generally the cause is unknown but associations can occur with alcohol use which is the only known dietary trigger of CH. There are other triggers such as strong odors (mainly solvents and cigarette smoke) and napping. CH is also characterized by unilateral, excruciating pain principally in ocular, frontal, and temporal areas, as well as ipsilateral lacrimation, conjunctival injection, photophobia, and nasal stuffiness. Attacks may happen at precise hours, especially at night.

The medical literature includes anecdotal reports of the use of 100% normobaric oxygen for the treatment of CH. Oxygen is an odorless, colorless gas at room temperature. It can be delivered in a chamber, by compressed air, via oxygen concentrator, or other method. Though often thought of as harmless, oxygen use has been noted to have adverse effects including blindness, pulmonary fibrosis, and suppression of the drive to breathe in patients who have advanced chronic obstructive lung disease. Oxygen is also known to increase fire risk in certain environments. There are a number of drug treatments for CH, including but not limited to IV and sublingual sumatriptan. Effective prophylactic drugs include prednisone, lithium, Methysergide, ergotamine, sodium valproate, and verapamil. At present, there is no curative treatment.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for claims with dates of services on or after January 4, 2011, the Centers for Medicare & Medicaid Services (CMS) believes that the available evidence suggests that the home use of oxygen to treat CH is promising and supports further research under §1862(a)(1)(E) of the Social Security Act (the Act) through the Coverage With Study Participation (CSP) form of Coverage With Evidence Development (CED).

The home use of oxygen to treat CH is covered by Medicare only for beneficiaries with CH participating in an approved prospective clinical study comparing normobaric 100% oxygen (NBOT) with at least one clinically appropriate comparator for the treatment of CH. The clinical study must address one or more aspects of the following questions:

1. Prospectively, compared to individuals with cluster headache who do not receive NBOT, do Medicare beneficiaries with CH who receive NBOT have improved outcomes as

indicated by:

- a. Pain relief
- b. Time to pain relief
- c. Durability of pain relief

2. Prospectively, among Medicare beneficiaries with cluster headache, which method of oxygen delivery provides the most benefit as indicated by:

- a. Pain relief
- b. Time to pain relief
- c. Durability of pain relief

3. Prospectively, among Medicare beneficiaries with cluster headache, what other factors, if any, predict the patient's response to 100% oxygen therapy as indicated by:

- a. Pain relief
- b. Time to pain relief
- c. Durability of pain relief

Only those beneficiaries diagnosed with the condition of cluster headache are eligible for participation in a clinical study. CMS adopts the diagnostic criteria used by the International Headache Society to form a definitive diagnosis of CH. Therefore, the home use of oxygen to treat CH is covered by Medicare only when furnished to Medicare beneficiaries who have had at least five severe to very severe unilateral headache attacks lasting 15-180 minutes when untreated. (Intensity of pain: Degree of pain usually expressed in terms of its functional consequence and scored on a verbal 5-point scale: 0=no pain; 1=mild pain, does not interfere with usual activities; 2=moderate pain, inhibits but does not wholly prevent usual activities; 3=severe pain, prevents all activities; 4=very severe pain. It may also be expressed on a visual analogue scale. <http://img.medscape.com/article/742/580/VAS.pdf>.)

The headaches must be accompanied by at least one of the following findings:

1. ipsilateral conjunctival injection and/or lacrimation; or
2. ipsilateral nasal congestion and/or rhinorrhea; or
3. ipsilateral eyelid edema; or
4. ipsilateral forehead and facial sweating; or
5. ipsilateral miosis and/or ptosis; or
6. a sense of restlessness or agitation.

The clinical study must adhere to the following standards of scientific integrity and relevance to the Medicare population:

- a. The principal purpose of the research study is to test whether a particular intervention potentially improves the participants' health outcomes.

- b. The research study is well supported by available scientific and medical information or it is intended to clarify or establish the health outcomes of interventions already in common clinical use.
- c. The research study does not unjustifiably duplicate existing studies.
- d. The research study design is appropriate to answer the research question being asked in the study.
- e. The research study is sponsored by an organization or individual capable of executing the proposed study successfully.
- f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found at 45 CFR Part 46. If a study is regulated by the Food and Drug Administration (FDA), it must be in compliance with 21 CFR parts 50 and 56.
- g. All aspects of the research study are conducted according to appropriate standards of scientific integrity (see <http://www.icmje.org>).
- h. The research study has a written protocol that clearly addresses, or incorporates by reference, the standards listed here as Medicare requirements for CED coverage.
- i. The clinical research study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Trials of all medical technologies measuring therapeutic outcomes as one of the objectives meet this standard only if the disease or condition being studied is life threatening as defined in 21 CFR § 312.81(a) and the patient has no other viable treatment options.
- j. The clinical research study is registered on the ClinicalTrials.gov website by the principal sponsor/investigator prior to the enrollment of the first study subject.
- k. The research study protocol specifies the method and timing of public release of all prespecified outcomes to be measured including release of outcomes if outcomes are negative or study is terminated early. The results must be made public within 24 months of the end of data collection. If a report is planned to be published in a peer reviewed journal, then that initial release may be an abstract that meets the requirements of the International Committee of Medical Journal Editors (<http://www.icmje.org>). However a full report of the outcomes must be made public no later than three (3) years after the end of data collection.
- l. The research study protocol must explicitly discuss subpopulations affected by the treatment under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria effect enrollment of these populations, and a plan for the retention and reporting of said populations on the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary.
- m. The research study protocol explicitly discusses how the results are or are not expected to be generalizable to the Medicare population to infer whether Medicare patients may

benefit from the intervention. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

Consistent with section 1142 of the Social Security Act, AHRQ supports clinical research studies that CMS determines meet the above-listed standards and address the above-listed research questions.

C. Nationally Non-Covered Indications

Effective for claims with dates of service on and after January 4, 2011, CMS believes that the evidence does not demonstrate that the home use of oxygen to treat CH improves health outcomes in Medicare beneficiaries with CH. Therefore, the home use of oxygen to treat CH is not reasonable and necessary under 1862(a)(1)(A) of the Act unless provided in the context of an approved clinical study under CED (see section B. above).

D. Other

This decision does not modify the existing requirements for coverage of the home use of oxygen currently identified in sections [240.2](#) and [240.2.1](#) of this manual. Additionally, the scope of the decision does not include any consideration of hyperbaric oxygen for any indication.

(This NCD last reviewed January 2011.)

Claims Processing Instructions

TN 2465 (Medicare Claims Processing) [↗](#)

TN 11263 (Medicare Claims Processing) [↗](#)

TN 11429 (Medicare Claims Processing) [↗](#)

Transmittal Information

Transmittal Number

130

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/downloads/R130NCD.pdf> [↗](#)

Revision History

01/2011 - After careful reconsideration, effective for claims with dates of service on and after January 4, 2011, Medicare will allow for coverage of home use of oxygen to treat Medicare beneficiaries diagnosed with CH when beneficiaries are enrolled in clinical studies

that are approved by CMS for the purpose of gaining further evidence. Effective date: 01/04/2011. Implementation date: 02/15/2011. ([TN130](#)) (CR7235)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [First reconsideration for Home Use of Oxygen to Treat Cluster Headache \(CAG-00296R\)](#)
- [Second reconsideration for Home Use of Oxygen and Home Oxygen Use to Treat Cluster Headaches \(CAG-00296R2\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Home Oxygen Use to Treat Cluster Headache (CH) - RETIRED	3	04/10/2023 - N/A	View
Home Oxygen Use to Treat Cluster Headache (CH)	2	09/27/2021 - 04/10/2023	View
Home Oxygen Use to Treat Cluster Headache (CH)	1	01/04/2011 - 09/27/2021	You are here